38 An up quark in a hadron changes to a down quark.

The elementary charge is e.

What is the magnitude of the change in the charge of the hadron?

A $\frac{1}{3}e$ **B** $\frac{2}{3}e$ **C** e **D** zero

39 A nucleus of magnesium-23 decays to form a nucleus of sodium-23, emitting a β^+ particle and particle X.

What is particle X?

A an antineutrino

B an electron

C a neutron

D a neutrino

40 A nucleus of $^{238}_{92}U$ decays in stages by emitting α-particles and β^- particles, eventually forming a nucleus of $^{206}_{82}Pb$.

How many α-particles and how many β^- particles are emitted during the decay chain?

	α-particles	β^- particles
A	8	6
B	8	10
C	16	6
D	16	22